The Star-Height Hierarchy of Kleene Algebra with Complement Is Infinite

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Abstract

We study the star height of terms of Kleene algebra with complement, that is, regular expressions extended with complement, where terms are compared by their equational theory with respect to languages: two terms are equivalent if they denote the same language under every valuation of the variables by languages. We show that the star-height hierarchy is infinite in this setting. For every $h \geq 0$ we give a term with four variables and star height $h + 1$ that is not equivalent to any term of star height at most h, even if the latter may use arbitrarily many variables. This contrasts with the standard interpretation, where letters are interpreted as themselves: there, it has recently been announced that every regular language has generalized star height at most three, so the corresponding hierarchy collapses. The same separation holds over finite relational models, whose equational theory is incomparable with that over languages. The proof works in a finite relational semantics that captures the equational theory with respect to languages, and in finite cyclic relational models. On suitable chain-like relational structures parameterized by an integer, the witness terms define periodic patterns whose periods are values of irreducible polynomials of degree growing exponentially in the star height. Conversely, using parametric Presburger arithmetic, we show that every term of lower star height has eventual periods given by polynomials whose irreducible factors have strictly smaller degree. A Bézout argument then separates the two.

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1 Introduction

The star height of a regular expression is the maximal nesting depth of Kleene stars in it. Star height has been a driving question in automata theory since Eggan [6]: Dejean and Schützenberger showed that the star-height hierarchy of regular languages is infinite already over a binary alphabet [5], and the decidability of star height, settled by Hashiguchi [9] and revisited by Kirsten [11], is a landmark of the field. When the complement is admitted as an operator, the picture changes completely. Languages of *generalized* star height 0 are the star-free languages, characterized by Schützenberger's theorem [20], but it has been a long-standing open problem whether the generalized star height of regular languages is bounded (see, e.g., [17]). Very recently, OpenAI announced a proof that every regular language has generalized star height at most three [15]; thus the generalized star-height hierarchy of regular languages collapses at level three, while it is still unknown whether level two is needed at all.

In this paper we consider star height for *terms* of Kleene algebra with complement [14], *w.r.t.* their equational theory *w.r.t.* languages. Here a term is a regular expression over variables built from 0, 1, $+$, $;$, $-^*$ and complement $\overline{-}$, and two terms are language-equivalent if they denote the same language under every valuation that maps the variables to arbitrary languages (over arbitrary alphabets). Equational theories *w.r.t.* languages are classical for

Kleene algebra [12] and its extensions with intersection, converse, or top [1, 3, 18], and for Kleene algebra with complement they are captured by a finite relational semantics [14]. Language-equivalence is finer than equivalence under the standard interpretation, in which each variable denotes itself as a letter, so star-height lower bounds *w.r.t.* languages are weaker than lower bounds for generalized star height. Nonetheless, to the best of our knowledge, it has not been known whether complement collapses the star-height hierarchy of terms *w.r.t.* languages, even at level two.

Contribution. We show that the hierarchy is infinite. Let $\mathcal{H}_h$ be the class of operations on languages that are defined by terms over the four variables x, y, z, c of star height at most h.

▶ **Theorem 1** (Main theorem). *For every $h \in \mathbb{N}$, there is a term Q_h over the variables x, y, z, c with $\mathrm{sh}(Q_h) = h + 1$ such that Q_h is not language-equivalent to any term t (over any variables) with $\mathrm{sh}(t) \leq h$. Hence $\mathcal{H}_0 \subsetneq \mathcal{H}_1 \subsetneq \mathcal{H}_2 \subsetneq \cdots$. Moreover, Q_h is not equivalent to any such t over finite relational models (hence over* REL*) either (Theorem 20).*

Thus, in contrast to the collapse at level three under the standard interpretation [15], the hierarchy *w.r.t.* languages is infinite. The witness Q_h is fixed in advance and does not depend on the competing term; only the counter-model depends on it. The separation also holds for the finite relational models RSUB of [14] (Section 3). Since validity *w.r.t.* languages and *w.r.t.* relational models are incomparable, the relational part is not a consequence of the language part. Moreover, the equational theory of relational models does not have the finite model property in general, so separation over *finite* relational models is a priori stronger than over REL.

Technical overview. The whole proof rests on one invariant, which we explain first.

Setting. By [14], it suffices to work with relations on finite chains. We use chain models $\mathfrak{M}_a$ (Figure 1) with m phases and an integer parameter a. In $\mathfrak{M}_a$, a term only sees time differences, so it defines, in each phase, an ultimately periodic set of integers. Its period depends on a, and turns out to be a polynomial $P(a)$ (on residue classes).

The invariant. We measure a term by the largest degree of an *irreducible factor* of its period polynomial P.

- *Witnesses can make it large* (Section 4). One star suffices to turn periods $q_1, \ldots, q_m$ (one per phase) into the periods $\mathrm{lcm}_k\, q_k + pi$ (in phase i). For suitable a, this lcm is a product of m shifted copies divided by a constant, and the result is irreducible by Eisenstein's criterion. So each star multiplies the irreducible degree by m, and the witness Q_h has an irreducible factor of degree m^h.
- *Arbitrary terms keep it small* (Section 5). Boolean operations and composition only multiply periods, so they create no new irreducible factors. A star multiplies the period by the weights of a few short cycles; each such weight is a polynomial whose degree is at most the total degree d of the period before the star. So a star creates new irreducible factors of degree at most d, while d grows by a factor of at most $m + 1$. Hence a term t of star height h has only irreducible factors of degree at most $\|t\|(m+1)^{h-1}$.

Since $m^h > \|t\|(m+1)^{h-1}$ for large m, the period of t cannot be a multiple of the period of Q_h for large a (by a Bézout argument), so t and Q_h differ. Section 3 states this reduction precisely.

Technical ingredients. That cycle weights are polynomials in a follows from the quasi-polynomiality of optima in parametric Presburger arithmetic [2]. To apply it after nested stars, we prove that weights of walks in graphs labeled by ultimately periodic parametric

Presburger families are again parametric Presburger (Lemma 16). For finite relational models we use cyclic versions of $\mathfrak{M}_a$, where the star case is even simpler (Section D).

Related work. Besides the star-height results above, the dot-depth hierarchy of star-free languages is infinite [4, 21], which is the analogue of our result *inside* generalized star height 0. The model class RSUB and the complexity of the equational theory of Kleene algebra with complement *w.r.t.* languages are from [14]; it is based on Pratt's embedding of languages into relations [19]. Kleene star corresponds to the transitive closure operator, and our terms can be seen as formulas of transitive closure logic with three variables. For full transitive closure logic on finite structures, Grädel and McColm proved a hierarchy theorem for the nesting of transitive closure and negation by Ehrenfeucht–Fraïssé games [8]. In contrast, on ordered finite structures, every formula of transitive closure logic is equivalent to one with a single transitive closure, at the price of higher arity [10]. Since our models are chains, the bounded arity of terms is essential in Theorem 1. Expressive-power separations for relational calculi have also been obtained by Ehrenfeucht–Fraïssé games [13]; our argument is instead arithmetic. On the arithmetic side, we use parametric Presburger arithmetic [2], and Parikh's theorem [16] together with semilinearity of Presburger sets [7].

Organization. Section 2 fixes the syntax and the semantics. Section 3 defines the witnesses and reduces Theorem 1 to two propositions, which are proved in Sections 4 and 5. Section 6 concludes. Omitted proofs are in the appendix.

2 Preliminaries

We write $\mathbb{N}$ for the set of non-negative integers, $\mathbb{N}_+ \triangleq \mathbb{N} \setminus \{0\}$, and $[l, r] \triangleq \{i \in \mathbb{Z} \mid l \le i \le r\}$. For subsets X, Y of an abelian group we write $X + Y \triangleq \{\alpha + \beta \mid \alpha \in X, \beta \in Y\}$, and $P\mathbb{N} \triangleq \{Pk \mid k \in \mathbb{N}\}$.

Let $P, T \in \mathbb{N}_+$. A set $X \subseteq \mathbb{N}$ is *P-periodic from T* if $n \in X \Leftrightarrow n + P \in X$ for all $n \ge T$. We then call P an *period* of X and T a *preperiod*; X is *ultimately periodic* if it has an period. Clearly, if X is P-periodic from T, then so is $\mathbb{N} \setminus X$, and X is kP-periodic from T' for all $k \in \mathbb{N}_+$ and $T' \ge T$; every finite subset of $[0, T-1]$ is P-periodic from T.

▶ **Fact 2.** *Let $X, Y \subseteq \mathbb{N}$.*

1. *If X is P-periodic from T and $X \cap [T, T+q+P] = q\mathbb{N} \cap [T, T+q+P]$ for some $q \in \mathbb{N}_+$, then $q \mid P$.*
2. *If X and Y are P-periodic from T_X and T_Y, respectively, then $X + Y$ is P-periodic from $T_X + T_Y + P$.*

Both are elementary; see Section A.

2.1 Terms and star height

Let $\mathbf{V}$ be a countably infinite set of *variables*. *Terms* are given by the grammar

$$t, s \quad ::= \quad x \mid 0 \mid 1 \mid t + s \mid t\,;s \mid t^* \mid \bar{t} \qquad (x \in \mathbf{V}).$$

We write $t \cap s \triangleq \overline{\bar{t} + \bar{s}}$. The *star height* $\mathrm{sh}(t)$ is the maximal nesting depth of $-^*$ in t: $\mathrm{sh}(x) = \mathrm{sh}(0) = \mathrm{sh}(1) = 0$, $\mathrm{sh}(\bar{t}) = \mathrm{sh}(t)$, $\mathrm{sh}(t+s) = \mathrm{sh}(t\,;s) = \max(\mathrm{sh}(t), \mathrm{sh}(s))$, and $\mathrm{sh}(t^*) = \mathrm{sh}(t) + 1$. In particular, complement does not count, and $\mathrm{sh}(t \cap s) = \max(\mathrm{sh}(t), \mathrm{sh}(s))$. The *size* $\|t\|$ is the number of occurrences of variables and constants in t.

2.2 Language semantics and relational semantics

A *language valuation* $\mathfrak{v}$ maps each variable to a language over some alphabet X; it extends to terms by interpreting 0, 1, $+$, $;$, $-^*$ as $\emptyset$, $\{\varepsilon\}$, union, concatenation, and Kleene star, and $\overline{}$ as complement relative to X^*. Two terms t, s are *language-equivalent*, written $\mathsf{LANG} \models t = s$, if they denote the same language under every language valuation. For $h \in \mathbb{N}$, we let $\mathcal{H}_h$ be the set of equivalence classes, *w.r.t.* language-equivalence, of terms t over $\{x, y, z, c\}$ with $\mathrm{sh}(t) \le h$.

Relational models. A *relational valuation* on a set Ω maps each variable to a binary relation on Ω. Its semantics $[\![t]\!]^{\mathsf{rel}}_{\mathfrak{v}} \subseteq \Omega^2$ interprets 0, 1, $+$, $;$, $-^*$ as the empty relation, the identity, union, relational composition, and reflexive transitive closure, and $\overline{}$ as complement relative to Ω^2. We write $\mathsf{REL} \models t = s$ if $[\![t]\!]^{\mathsf{rel}}_{\mathfrak{v}} = [\![s]\!]^{\mathsf{rel}}_{\mathfrak{v}}$ for every relational valuation $\mathfrak{v}$ on every set Ω.

Forward semantics and RSUB. Let $(\Omega, \le)$ be a linear order in which every interval $[u, v] \triangleq \{w \mid u \le w \le v\}$ is finite. Its set of *forward pairs* is $U_\Omega \triangleq \{(u, v) \in \Omega^2 \mid u \le v\}$. For a relational valuation $\mathfrak{v}$ on Ω with $\mathfrak{v}(x) \subseteq U_\Omega$ for all x, the *forward semantics* $[\![t]\!]_{\mathfrak{v}} \subseteq U_\Omega$ is defined like $[\![t]\!]^{\mathsf{rel}}_{\mathfrak{v}}$, except that complement is taken relative to U_Ω: $[\![\overline{t}]\!]_{\mathfrak{v}} = U_\Omega \setminus [\![t]\!]_{\mathfrak{v}}$. Since all relations then consist of forward pairs, the intermediate points of a composition and of a closure path from u to v lie in $[u, v]$. An induction on t thus gives:

▶ **Lemma 3** (*Interval locality*). *For $u \le v$ in Ω, whether $(u, v) \in [\![t]\!]_{\mathfrak{v}}$ depends only on the isomorphism type of $([u, v], \le, (\mathfrak{v}(x) \cap [u, v]^2)_x)$.*

An RSUB *valuation* is a relational valuation $\mathfrak{v}$ on a finite chain $[0, n]$ with $\mathfrak{v}(x) \subseteq U_{[0,n]}$ such that, for each x, $\mathfrak{v}(x)$ contains all pairs (u, u) or none of them. We write $\mathsf{RSUB} \models t = s$ if $[\![t]\!]_{\mathfrak{v}} = [\![s]\!]_{\mathfrak{v}}$ for every RSUB valuation $\mathfrak{v}$.

▶ **Theorem 4** ([14, Thm. 3.2]). *For all terms t, s: $\mathsf{LANG} \models t = s$ if and only if $\mathsf{RSUB} \models t = s$.*

We say that $\mathfrak{v}$ is *diagonal-uniform* if each $\mathfrak{v}(x)$ contains all pairs (u, u) or none of them. By Lemma 3 and Theorem 4, a single discrepancy on an infinite chain suffices to separate two terms *w.r.t.* languages:

▶ **Corollary 5.** *Let $\mathfrak{v}$ be a diagonal-uniform relational valuation on Ω with $\mathfrak{v}(x) \subseteq U_\Omega$, and let $(u, v) \in [\![t]\!]_{\mathfrak{v}} \setminus [\![s]\!]_{\mathfrak{v}}$. Then $\mathsf{RSUB} \not\models t = s$; hence t and s are not language-equivalent.*

Proof. Restrict $\mathfrak{v}$ to the finite interval $[u, v]$ and rename it to a chain $[0, n]$. The result is an RSUB valuation since it is diagonal-uniform, and by Lemma 3 it still distinguishes t and s at the endpoints. ◀

Validity *w.r.t.* LANG and *w.r.t.* REL are incomparable: for instance, $x \le x\,;\overline{0}\,;x$ (*i.e.*, $x \cap \overline{x\,;\overline{0}\,;x} = 0$) holds in REL but not in LANG [18], and conversely there are equations valid in LANG but not in REL [14]. Hence neither separation implies the other. The main text treats language-equivalence; finite relational models are treated in Section D.

2.3 Chain models with phases

Throughout the paper, we fix an integer $m \ge 2$ and a prime $p > m$; constants hidden in $O(\cdot)$ may depend on m and p. Let $\Omega \triangleq \mathbb{N} \times [1, m]$, ordered lexicographically: $(t, i) < (t', i')$ iff

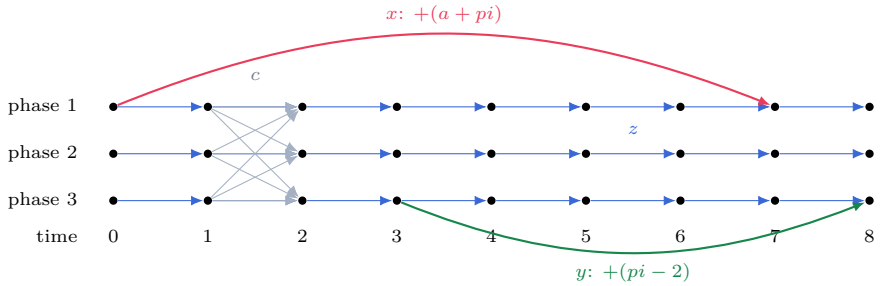


Figure 1 The chain model $\mathfrak{M}_a$ for $m = 3$ (schematic; edge lengths are not to scale). Points are ordered lexicographically by time and phase. z relates points of the same phase with non-negative time difference (only steps of length 1 are drawn), and c moves one time step between arbitrary phases (shown only between times 1 and 2). x and y jump within the phase i by $a + pi$ and $pi - 2$ time steps, respectively.

$t < t'$, or $t = t'$ and $i < i'$. We call t the *time* and i the *phase* of (t, i). For $a \in \mathbb{N}_+$, the *chain model* $\mathfrak{M}_a$ is the relational valuation on Ω with

$$z \mapsto \{((t,i),(t+k,i)) \mid k \in \mathbb{N}\}, \qquad x \mapsto \{((t,i),(t+a+pi,i))\},$$
$$c \mapsto \{((t,i),(t+1,j))\}, \qquad y \mapsto \{((t,i),(t+pi-2,i))\},$$

where t ranges over $\mathbb{N}$ and i, j over $[1, m]$, and all other variables are mapped to $\emptyset$; we use the forward semantics. Thus z relates two points of the same phase with non-negative time difference, and c moves one time step between arbitrary phases. All these pairs are forward pairs; $\mathfrak{M}_a(z)$ contains all pairs (u, u), and the other variables contain none, so $\mathfrak{M}_a$ is diagonal-uniform. Figure 1 depicts $\mathfrak{M}_a$.

For a term t and phases i, j, we define the *difference set*

$$L^t_{ij}(a) \triangleq \{n \in \mathbb{N} \mid ((0,i),(n,j)) \in [\![t]\!]_{\mathfrak{M}_a}\}.$$

(Note that $0 \notin L^t_{ij}(a)$ if $j < i$, as $((0,i),(0,j))$ is not a forward pair.) The shift $(t, i) \mapsto (t+s, i)$ maps each interval of Ω isomorphically onto another one and preserves $\mathfrak{M}_a$. Hence, by Lemma 3, the truth value of t at a pair depends only on the two phases and the time difference:

$$((t,i),(t',j)) \in [\![t]\!]_{\mathfrak{M}_a} \iff t' - t \in L^t_{ij}(a). \tag{1}$$

3 Proof strategy

By Theorem 4 and Corollary 5, to prove the language part of Theorem 1 it suffices to show the following: *for every h and every term t with $\mathrm{sh}(t) \le h$, there are m, p, and a such that $L^t_{11}(a) \neq L^{Q_h}_{11}(a)$.* This follows from two propositions about the chain models $\mathfrak{M}_a$.

The witness terms. For a term Q, we define the terms $\mathsf{M}(Q)$, $\mathsf{J}(Q)$, and $\Phi(Q)$ by

$$\mathsf{M}(Q) \triangleq (z \cap \overline{1 + c + c\,;\,c}) \cap \overline{c\,;\,(z \cap \overline{Q})\,;\,c}, \qquad \mathsf{J}(Q) \triangleq \mathsf{M}(Q) \cap \overline{\mathsf{M}(Q)\,;\,c\,;\,z},$$
$$\Phi(Q) \triangleq (\mathsf{J}(Q)\,;\,y)^*,$$

and let $Q_0 \triangleq x^*$ and $Q_{h+1} \triangleq \Phi(Q_h)$. Then $\mathrm{sh}(Q_h) = h + 1$, and Q_h is a fixed term over x, y, z, c, independent of m, p, and a. Intuitively, if Q expresses "the time difference is a

multiple of q_i" in each phase i, then $\mathsf{M}(Q)$ expresses "the time difference is $2+rL$ with $r \geq 1$" for $L = \operatorname{lcm}_i q_i$, $\mathsf{J}(Q)$ keeps the first such jump, and $\Phi(Q)$ expresses "the time difference is a multiple of $L + pi$". All complements occur inside an intersection with z; this makes the same terms work in finite cyclic models (Section D).

Two propositions. Let $H \triangleq \operatorname{lcm}(1, \ldots, m)$ and $\mathcal{A} \triangleq \{a \in \mathbb{N}_+ \mid a \equiv H \pmod{pH}\}$.

▶ **Proposition 6** (Witness; Section 4). *For each $h \in \mathbb{N}$ there is an irreducible polynomial $N_h \in \mathbb{Z}[a]$ of degree m^h such that, for every $a \in \mathcal{A}$,*

$$L_{11}^{Q_h}(a) = q_h(a)\mathbb{N} \quad \text{for some integer } q_h(a) \geq a \text{ dividing } N_h(a).$$

▶ **Proposition 7** (Periods; Section 5). *Let t be a term with $\operatorname{sh}(t) \leq h$, and let $\beta \triangleq \|t\|(m+1)^{h-1}$ if $h \geq 1$ and $\beta \triangleq 0$ if $h = 0$. There are $M \in \mathbb{N}_+$ and $P_0, \ldots, P_{M-1} \in \mathbb{Q}[a]$, all of whose irreducible factors have degree at most β, such that for each r and all large $a \equiv r \pmod{M}$, the set $L_{11}^t(a)$ is ultimately periodic with period $P_r(a)$.*

In other words, the periods of terms of low star height are given by a quasi-polynomial in a whose constituents have only irreducible factors of low degree, whereas the witness needs an irreducible factor of degree m^h.

Proof of Theorem 1 (language part). Let $\operatorname{sh}(t) \leq h$. Choose $m \geq 2$ with $m^h > \beta$ (possible since $m^h/(m+1)^{h-1} \geq m/2^{h-1}$) and a prime $p > m$. Take M and P_r from Proposition 7, where r is a residue with infinitely many $a \in \mathcal{A}$, $a \equiv r \pmod{M}$; write $P \triangleq P_r$. Suppose, for contradiction, that $L_{11}^t(a) = L_{11}^{Q_h}(a) = q_h(a)\mathbb{N}$ for all these a. A period of $q_h(a)\mathbb{N}$ is a multiple of $q_h(a)$, so $q_h(a) \mid P(a)$. Since N_h is irreducible of degree $m^h > \beta$, it is coprime to P in $\mathbb{Q}[a]$. Clearing denominators in a Bézout identity gives $U, V \in \mathbb{Z}[a]$ and $K \in \mathbb{N}_+$ with $UN_h + VP = K$. As $q_h(a)$ divides the integers $N_h(a)$ and $P(a)$, it divides K; this is impossible for $a > K$, since $q_h(a) \geq a$. So $L_{11}^t(a) \neq L_{11}^{Q_h}(a)$ for some a, and t is not language-equivalent to Q_h. Finally, $\operatorname{sh}(Q_h) = h+1$, so the class of Q_h lies in $\mathcal{H}_{h+1} \setminus \mathcal{H}_h$. ◀

The relational part is proved in the same way with finite cyclic versions of $\mathfrak{M}_a$ (Section D). Note the order of quantifiers: the witness Q_h is fixed, while m, p, and a (hence the counter-model) depend on t.

4 The witness

In this section we prove Proposition 6.

4.1 An operator that turns an lcm into a period

▶ **Lemma 8.** *Let $q_1, \ldots, q_m \in \mathbb{N}_+$ and $L \triangleq \operatorname{lcm}(q_1, \ldots, q_m)$. If $L_{ii}^Q(a) = q_i\mathbb{N}$ for all i, then $L_{ii}^{\Phi(Q)}(a) = (L + pi)\mathbb{N}$ for all i.*

Proof. We use (1) throughout. The term $1 + c + c\,;c$ relates exactly the pairs with time difference 0, 1, or 2 (in any phases), so $z \cap \overline{1 + c + c\,;c}$ relates exactly the pairs in the same phase with time difference at least 3. The term $z \cap \overline{Q}$ relates (t,k) and $(t+n,k)$ iff $q_k \nmid n$; the values of Q across different phases play no role. Consider a pair $((t,i),(t+n,i))$ with $n \geq 3$. As c moves one time step between arbitrary phases, the pair is in $c\,;(z \cap \overline{Q})\,;c$ iff $q_k \nmid n-2$ for some k, *i.e.*, iff $L \nmid n-2$. Therefore $\mathsf{M}(Q)$ relates exactly the pairs $((t,i),(t+2+rL,i))$ with $r \geq 1$. The term $c\,;z$ relates exactly the pairs whose time strictly

increases, so $\mathsf{M}(Q)\,;c\,;z$ relates u to v iff some $\mathsf{M}(Q)$-successor of u has smaller time than v. So $\mathsf{J}(Q)$ keeps only the first $\mathsf{M}(Q)$-successor, and relates exactly the pairs $((t,i),(t+L+2,i))$. Then $\mathsf{J}(Q)\,;y$ relates exactly $((t,i),(t+L+pi,i))$, and its reflexive transitive closure gives $L_{ii}^{\Phi(Q)}(a) = (L+pi)\mathbb{N}$. ◀

Let $f_0(a) \triangleq a$ and $f_{r+1}(a) \triangleq \mathrm{lcm}_{i\in[1,m]}(f_r(a)+pi)$, and let $q_r(a) \triangleq f_r(a)+p$. Since $L_{ii}^{x^*}(a) = (a+pi)\mathbb{N}$, induction with Lemma 8 yields:

▶ **Corollary 9.** *$L_{ii}^{Q_r}(a) = (f_r(a)+pi)\mathbb{N}$ for all r, i, and a. In particular, $L_{11}^{Q_r}(a) = q_r(a)\mathbb{N}$.*

4.2 The periods are values of irreducible polynomials

Recall $H = \mathrm{lcm}(1,\dots,m)$, and let $\Delta \triangleq m!/H$; note that $p \nmid H\Delta$ as $p > m$. The lcm in the definition of f_r is a product divided by Δ under the following condition.

▶ **Lemma 10.** *Let $b \in \mathbb{N}_+$ with $H \mid b$ and $p \nmid b$. Then $\mathrm{lcm}_{i\in[1,m]}(b+pi) = \frac{1}{\Delta}\prod_{i=1}^{m}(b+pi)$, and this number is divisible by H but not by p.*

The proof (Section A) compares ℓ-adic valuations for each prime ℓ. Every a in the progression $\mathcal{A}$ of Section 3 satisfies $H \mid a$ and $p \nmid a$, so Lemma 10 applies inductively, and $f_{r+1}(a) = \frac{1}{\Delta}\prod_{i=1}^{m}(f_r(a)+pi)$ for all $a \in \mathcal{A}$. To clear denominators, let $\delta_r \triangleq (m^r-1)/(m-1)$ and define $A_r, N_r \in \mathbb{Z}[a]$ by

$$A_0(a) \triangleq a, \qquad A_{r+1}(a) \triangleq \prod_{i=1}^{m}\left(A_r(a)+pi\Delta^{\delta_r}\right), \qquad N_r(a) \triangleq A_r(a)+p\Delta^{\delta_r}.$$

As $\delta_{r+1} = 1+m\delta_r$, induction on r gives $f_r(a) = A_r(a)/\Delta^{\delta_r}$ and hence

$$q_r(a) = N_r(a)/\Delta^{\delta_r} \qquad \text{for all } a \in \mathcal{A}. \tag{2}$$

▶ **Lemma 11.** *For every $r \in \mathbb{N}$, N_r is a monic polynomial of degree m^r that is irreducible over $\mathbb{Q}$.*

Proof. By induction, A_r is monic of degree m^r, and so is N_r. For $r=0$, $N_0(a) = a+p$ is linear. Let $r \geq 1$; we apply Eisenstein's criterion at p. Modulo p, we have $A_{r+1} \equiv {A_r}^m$, so $A_r \equiv a^{m^r}$ and $N_r \equiv a^{m^r} \pmod p$. Thus all non-leading coefficients of N_r are divisible by p. For the constant term, p divides every factor $A_{r-1}(0)+pi\Delta^{\delta_{r-1}}$ (as $A_0(0)=0$ and $A_{r-1}(0) \equiv 0 \pmod p$), so $p^m \mid A_r(0)$ with $m \geq 2$. Since $p \nmid \Delta$, we get $N_r(0) \equiv p\Delta^{\delta_r} \not\equiv 0 \pmod{p^2}$. ◀

Proof of Proposition 6. By Corollary 9, $L_{11}^{Q_h}(a) = q_h(a)\mathbb{N}$ with $q_h(a) \geq a$, and $q_h(a)$ divides $N_h(a) = \Delta^{\delta_h}q_h(a)$ by (2). Lemma 11 gives irreducibility and the degree. ◀

Only the *irreducibility* of N_h is used; we need no information on the prime factorization of the values $q_h(a)$.

5 Periods of terms of bounded star height

In this section we prove Proposition 7. Let $D_{ij} \triangleq \mathbb{N}$ if $i \leq j$ and $D_{ij} \triangleq \mathbb{N}_+$ if $j < i$; these are the possible time differences from phase i to phase j.

term	d_-	b_-	period polynomial P_-
variable or constant	1	0	1
$\overline{s}$	d_s	b_s	P_s
$s+u$, $s\,;u$	d_s+d_u	$\max(b_s,b_u)$	$P_s\cdot P_u$
s^*	$(m+1)d_s$	d_s	$P_s\cdot\prod_C w_C$

Table 1 The degree bound d_t of the period polynomial and the bound b_t on the degrees of its irreducible factors (Claim 13). The product in the last row ranges over the cyclic strongly connected components C of the phase graph of s, and w_C is the least positive weight of a short closed walk in C (Section 5.1).

Parametric Presburger arithmetic. Following [2], a *parametric Presburger family* is a family $(S_a)_{a\in\mathbb{N}}$ of sets $S_a\subseteq\mathbb{Z}^k$ defined by a first-order formula over $(\mathbb{Z},+,\le)$ in which a variable may be multiplied by a polynomial in the parameter a (but two variables may not be multiplied, and a is not quantified). A function $g\colon\mathbb{N}\to\mathbb{Z}$ is a *quasi-polynomial* if there are $M\in\mathbb{N}_+$ and $g_0,\dots,g_{M-1}\in\mathbb{Q}[a]$ with $g(a)=g_r(a)$ for all large $a\equiv r \pmod M$.

▶ **Theorem 12** ([2, Thm. 1.15, Cor. 1.16]). *If $(S_a)_a$ is a parametric Presburger family with $S_a\subseteq\mathbb{N}$, then $\{a\mid S_a\neq\emptyset\}$ is ultimately periodic, and $a\mapsto\min S_a$ agrees with a quasi-polynomial whenever $S_a\neq\emptyset$.*

The inductive claim. We define $d_t\in\mathbb{N}_+$ and $b_t\in\mathbb{N}$ by induction on t as in Table 1; by induction, $b_t\le\|t\|(m+1)^{\mathrm{sh}\,(t)-1}$ if $\mathrm{sh}\,(t)\ge1$ and $b_t=0$ otherwise. Proposition 7 follows from the following stronger claim (with $\beta\ge b_t$).

▷ Claim 13. For every term t:

1. each $(L^t_{ij}(a))_a$ is a parametric Presburger family; and
2. there are $M\in\mathbb{N}_+$, $C\in\mathbb{N}$, and, for each residue r modulo M, a polynomial $P_r\in\mathbb{Q}[a]$ of degree at most d_t whose irreducible factors have degree at most b_t, such that for all large $a\equiv r \pmod M$, every $L^t_{ij}(a)$ is $P_r(a)$-periodic from Ca^{d_t}.

Item 1 and the polynomial bound on the preperiod are only needed to keep the induction going.

5.1 Proof of the claim

We proceed by induction on t. Refining residue classes is harmless: if t satisfies Item 2 with modulus M, it does so with any multiple of M. So, when combining finitely many terms, we may assume that they share the modulus, and we fix one residue class r. For a term t, we write P_t for its period polynomial on this class, and $T_t(a)\triangleq Ca^{d_t}$ for its preperiod. Since a polynomial of degree at most d is $O(a^d)$, we freely absorb sums and products of such bounds into Ca^d.

Variables and constants. Each difference set is $\emptyset$, $\{0\}$, $\{1\}$, $\{pi-2\}$, $\{a+pi\}$, or $\mathbb{N}$ (for z). Each is clearly a parametric Presburger family and is 1-periodic from $a+pm+1=O(a)$. So we can take $P_t=1$, which is consistent with $d_t=1$ and $b_t=0$.

Complement and union. We have $L^{\bar t}_{ij}(a)=D_{ij}\setminus L^t_{ij}(a)$, and D_{ij} is 1-periodic from 1. Hence P_t remains an period, with preperiod $T_t(a)+1$. For union, $L^{t+s}_{ij}(a)=L^t_{ij}(a)\cup L^s_{ij}(a)$ is $P_t(a)P_s(a)$-periodic from $\max(T_t(a),T_s(a))$. Definability is preserved in both cases.

Composition. By (1), the intermediate point of a composition can be in any phase k:

$$L_{ij}^{t;s}(a) = \bigcup_{k=1}^{m} \left(L_{ik}^{t}(a) + L_{kj}^{s}(a)\right).$$

(The order constraints on the intermediate point are already encoded in the difference sets.) This is again a parametric Presburger family. Let $P \triangleq P_t P_s$. All sets on the right-hand side are $P(a)$-periodic, so by Fact 2(2) the union is $P(a)$-periodic from $T_t(a) + T_s(a) + P(a) = O(a^{d_t+d_s})$. We stress that a *single* period polynomial is used for all pairs of phases; therefore the union over the intermediate phases does not increase the degree of the period polynomial further.

Kleene star. This is the only case where new irreducible factors appear. Let F satisfy Claim 13, with $d \triangleq d_F$, $b \triangleq b_F \le d$, period P_F and preperiod T_F on our residue class. The *phase graph* of F at a has vertex set $[1, m]$, an edge $k \to l$ iff $S_{kl}(a) \triangleq L_{kl}^{F}(a) \neq \emptyset$, and edge weight sets $S_{kl}(a)$. A walk $k_0 \to k_1 \to \cdots \to k_\ell$ has *weight* $s_1 + \cdots + s_\ell$ for any choice of $s_e \in S_{k_{e-1}k_e}(a)$. Since F^* is the reflexive transitive closure, (1) gives

$$L_{ij}^{F^*}(a) = \{\text{weights of walks from } i \text{ to } j \text{ in the phase graph of } F \text{ at } a\}. \tag{3}$$

For a strongly connected component C of the phase graph, let $G_C(a) \subseteq \mathbb{N}$ be the set of weights of closed walks of positive length inside C. We call an element of an edge set *small* if it is at most $B \triangleq T_F + 2P_F$.

▶ **Lemma 14.** *Let $S_{kl} \subseteq \mathbb{N}$ be P-periodic from T, and let $B \triangleq T + 2P$. For every strongly connected component C, every element of G_C is an integer linear combination of weights of closed walks inside C of length at most m all of whose chosen elements are at most B. In particular, if $G_C \not\subseteq \{0\}$, then C has a closed walk of length at most m whose weight lies in $[1, mB]$.*

Step 1: cycle weights. For each strongly connected component C, the family of weights of closed walks of length between 1 and m inside C is a finite union of sums of the families S_{kl}, hence a parametric Presburger family. By Theorem 12, after refining the modulus, on our residue class (for large a) the phase graph does not depend on a, the condition $G_C(a) \not\subseteq \{0\}$ does not depend on a (by Lemma 14), and

$$w_C(a) \triangleq \min\{w \ge 1 \mid w \text{ is the weight of a closed walk inside } C \text{ of length at most } m\}$$

is a polynomial in a for each C with $G_C(a) \not\subseteq \{0\}$. By Lemma 14, $w_C(a) \le mB(a) = O(a^d)$, hence $\deg w_C \le d$. We let

$$P_{F^*} \triangleq P_F \cdot \prod_{C\,:\,G_C \not\subseteq \{0\}} w_C.$$

There are at most m components, so $\deg P_{F^*} \le (m+1)d$, and every irreducible factor of P_{F^*} divides P_F or some w_C, so it has degree at most d.

Step 2: periodicity of walk weights.

▶ **Lemma 15.** *Let $S_{kl} \subseteq \mathbb{N}$ ($k, l \in [1, m]$) be P-periodic from T, and let $B \triangleq T + 2P$. For each strongly connected component C with $G_C \not\subseteq \{0\}$, let $w_C \in G_C \setminus \{0\}$, and let $\Pi \triangleq P \prod_C w_C$. Then for all i, j, the set of weights of walks from i to j is Π-periodic from $c_m(B^2 + \Pi)$, where c_m depends only on m.*

The idea of the proof (Section B) is as follows. A walk passes through a sequence of strongly connected components, so its weight set is a finite union of sums of edge sets and of the sets of weights of walks from e to f inside a component C. The closed walks at e inside C form a submonoid of $\mathbb{N}$ whose gcd g_C divides w_C. By Lemma 14, this monoid contains a set of elements of size $O(mB)$ with gcd g_C, so a Frobenius-type bound shows that it contains all multiples of g_C from $O(B^2)$ on. Hence the walks from e to f inside C realize, from $O(B^2)$ on, exactly one residue class modulo g_C. Finally, Fact 2(2) combines the at most $2m$ summands.

Applied with $P = P_F(a)$, $T = T_F(a)$, and $w_C = w_C(a)$, Lemma 15 shows that $L_{ij}^{F^*}(a)$ is $P_{F^*}(a)$-periodic from $O(a^{2d} + a^{(m+1)d}) = O(a^{(m+1)d})$.

Step 3: definability. It remains to show that $(L_{ij}^{F^*}(a))_a$ is a parametric Presburger family. Kleene star is not available in parametric Presburger arithmetic, and quasi-polynomiality of optima alone does not imply this closure property. By (3), it follows from the next lemma, whose hypotheses are exactly provided by Claim 13 for F.

▶ **Lemma 16** (Closure lemma). *Let $(S_{kl}(a))_a$ for $k, l \in [1, m]$ be parametric Presburger families of subsets of $\mathbb{N}$. Assume there are $M \in \mathbb{N}_+$ and, for each residue r modulo M, polynomials $P_r, T_r \in \mathbb{Q}[a]$ such that for all sufficiently large $a \equiv r \pmod M$, every $S_{kl}(a)$ is $P_r(a)$-periodic from $T_r(a)$. Then for all i, j, the family of the sets of weights of walks from i to j (with edge weight sets $S_{kl}(a)$) is a parametric Presburger family.*

We sketch the proof; the details are in Section C.

- *Bounded sets have a simplex normal form.* Let $(X_a)_a$ be a parametric Presburger family with $X_a \subseteq [0, Ca^D]$. Using the normalization to polynomially bounded quantifiers [2, Thm. 3.7] and writing every variable in base a (cf. [2, Sect. 4.3]), the set of pairs $(a, \text{digits of } n)$ with $n \in X_a$ becomes definable in ordinary Presburger arithmetic, hence semilinear [7]. Since digits are smaller than a, every period vector of this semilinear set increases the a-coordinate. This yields that, for large a, X_a is a finite union of sets of the form

$$A(a) + \Big\{ \textstyle\sum_j \lambda_j W_j(a) \mid \lambda \in \mathbb{N}^J,\ \sum_j \lambda_j = K(a) \Big\}, \tag{4}$$

  where $A, W_j \in \mathbb{Z}[a]$ and K is an affine function of a.
- *Iterated sums stay Presburger.* The sum of k elements of the set (4) is exactly $kA(a) + \{\sum_j \lambda_j W_j(a) \mid \sum_j \lambda_j = kK(a)\}$, since a vector with coordinate sum $kK(a)$ splits into k vectors with coordinate sum $K(a)$. Here only products of a polynomial in a and a variable occur.
- *Walks.* Each edge set splits into a bounded part below $T_r(a)$ and a tail (one window of length $P_r(a)$, shifted by $P_r(a)\mathbb{N}$). The vectors counting how often each part of each edge is used along walks from i to j form the Parikh image of a regular language, which is semilinear [16]. Combining it with the previous item gives a parametric Presburger definition of the walk weights.

This completes the Kleene star case. The resulting bounds are $d_{F^*} = (m+1)d$ and $b_{F^*} = d$, as in Table 1, which proves Claim 13. ◀

6 Conclusion

We have shown that the star-height hierarchy of Kleene algebra with complement is infinite *w.r.t.* languages, already for terms over four variables (Theorem 1), and also over the finite relational models RSUB and over finite relational models. The separating invariant is arithmetic: the degree of the irreducible factors of the polynomial that describes the

periods of a term in parameterized chain models. To propagate this invariant through nested stars, we have shown that the weights of walks in graphs labeled by ultimately periodic parametric Presburger families are again parametric Presburger (Lemma 16), which may be of independent interest.

We conclude with some open questions.

- *Fewer variables.* Does the hierarchy remain infinite for terms with fewer than four variables, in particular with a single variable?
- *Restricted model classes.* Our counter-models have unbounded treewidth. Is the hierarchy still infinite over structures of bounded treewidth, *e.g.*, over series-parallel structures?
- *Standard interpretation.* Under the standard interpretation, every regular language has generalized star height at most three [15], and it remains open whether height two is ever needed [17]. Which intermediate semantics, between arbitrary languages and letters, still yield an infinite hierarchy?
- *Decidability.* Is it decidable, given a term t and $h \in \mathbb{N}$, whether t is language-equivalent to a term of star height at most h? Since language-equivalence itself is Π_1^0-complete [14], the exact complexity of this problem is an interesting question; already the case $h = 0$ (definability by a star-free term) is open to our knowledge.

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A Omitted proofs for the preliminaries and the witness

▶ **Fact 2.** *Let $X, Y \subseteq \mathbb{N}$.*

1. *If X is P-periodic from T and $X \cap [T, T+q+P] = q\mathbb{N} \cap [T, T+q+P]$ for some $q \in \mathbb{N}_+$, then $q \mid P$.*
2. *If X and Y are P-periodic from T_X and T_Y, respectively, then $X+Y$ is P-periodic from $T_X + T_Y + P$.*

Proof. (1) Pick $n \in [T, T+q-1]$ with $q \mid n$. Then $n \in X$, so $n+P \in X \cap [T, T+q+P] \subseteq q\mathbb{N}$, whence $q \mid P$.

(2) Let $n \geq T_X + T_Y + P$. If $n = \alpha + \beta$ with $\alpha \in X$, $\beta \in Y$, then $\alpha \geq T_X$ or $\beta \geq T_Y$; say $\alpha \geq T_X$. Then $\alpha + P \in X$, so $n + P \in X + Y$. Conversely, if $n + P = \alpha + \beta$, then $\alpha \geq T_X + P$ or $\beta \geq T_Y + P$, since otherwise $\alpha + \beta \leq T_X + T_Y + 2P - 2 < n + P$; say $\alpha \geq T_X + P$. Then $\alpha - P \geq T_X$ and $\alpha - P \in X$, so $n \in X + Y$. ◀

Proof of Lemma 3. By induction on t, we show the claim simultaneously for all $u \leq v$. The cases of variables, constants, union, and complement are immediate (in the forward semantics, complement is taken relative to forward pairs, and (u, v) is one). If $(u, v) \in [\![t\,;s]\!]_{\mathfrak{v}}$ via w, then $u \leq w \leq v$, and we apply the induction hypothesis to (u, w) and (w, v), whose intervals are contained in $[u, v]$. The case of t^* is similar, since all points of a closure path from u to v lie in $[u, v]$. The statement on isomorphism types follows in the same way. ◀

▶ **Lemma 10.** *Let $b \in \mathbb{N}_+$ with $H \mid b$ and $p \nmid b$. Then $\mathrm{lcm}_{i\in[1,m]}(b+pi) = \frac{1}{\Delta}\prod_{i=1}^{m}(b+pi)$, and this number is divisible by H but not by p.*

Proof. Let $\Pi \triangleq \prod_{i=1}^{m}(b+pi)$ and $\Lambda \triangleq \mathrm{lcm}_i(b+pi)$. We compare ℓ-adic valuations v_ℓ for each prime ℓ.

Case $\ell = p$. No factor is divisible by p, since $p \mid b + pi$ would imply $p \mid b$. As $p > m$, also $p \nmid \Delta$.

Case $\ell > m$, $\ell \neq p$. If ℓ divides $b+pi$ and $b+pi'$, then $\ell \mid p(i-i')$, so $\ell \mid i-i'$, whence $i = i'$ as $|i-i'| < m < \ell$. So at most one factor is divisible by ℓ, and $v_\ell(\Pi) = v_\ell(\Lambda)$. Also $v_\ell(\Delta) = 0$.

Case $\ell \leq m$. Let $e \triangleq \lfloor \log_\ell m \rfloor$, so that $v_\ell(H) = e$, and $\ell^e \mid b$. For $s \leq e$, we have $\ell^s \mid b+pi$ iff $\ell^s \mid i$, which holds for exactly $\lfloor m/\ell^s \rfloor \geq 1$ indices i. For $s > e$, at most one index i satisfies $\ell^s \mid b+pi$, by the argument of the previous case with $\ell^s > m$. Let σ be the number of $s > e$ for which such an index exists; since the sets $\{i \mid \ell^s \mid b+pi\}$ decrease in s, $v_\ell(\Lambda) = \max_i v_\ell(b+pi) = e+\sigma$. On the other hand, $v_\ell(\Pi) = \sum_{s\geq 1} \#\{i \mid \ell^s \mid b+pi\} = \sum_{s=1}^{e} \lfloor m/\ell^s \rfloor + \sigma = v_\ell(m!)+\sigma$ by Legendre's formula. Hence $v_\ell(\Pi) - v_\ell(\Lambda) = v_\ell(m!) - v_\ell(H) = v_\ell(\Delta)$, and $v_\ell(\Lambda) \geq e = v_\ell(H)$.

Together, $\Pi = \Delta\Lambda$, $H \mid \Lambda$, and $p \nmid \Lambda$. ◀

B Omitted proofs for the period bound

Proof of the bounds on d_t and b_t. By induction on t, with $h \triangleq \mathrm{sh}(t)$. For variables and constants, $b_t = 0 < 1 = d_t = \|t\|$. Complement changes nothing. For $t = t+s$ or $t\,;s$: $d_t = d_t + d_s \leq (\|t\|+\|s\|)(m+1)^h = \|t\|(m+1)^h$, and $b_t = \max(b_t, b_s)$ is 0 if $h = 0$ and at most $\max(\|t\|,\|s\|)(m+1)^{h-1} \leq \|t\|(m+1)^{h-1}$ otherwise (summands of star height 0 contribute 0). For $t = t^*$: $d_t = (m+1)d_t \leq \|t\|(m+1)^h$ and $b_t = d_t \leq \|t\|(m+1)^{h-1}$, as $\mathrm{sh}(t) = h-1$. Finally, $b_t \leq d_t$ is preserved by each rule. ◀

Throughout the rest of this section, $S_{kl} \subseteq \mathbb{N}$ $(k,l \in [1,m])$ are P-periodic from T, and $B \triangleq T+2P$. Walks, weights, and strongly connected components refer to the graph with an edge $k \to l$ iff $S_{kl} \neq \emptyset$. An element $s \in S_{kl}$ is small if $s \leq B$. If $s \in S_{kl}$ is not small, let $s' \in [T, T+P-1]$ with $s' \equiv s \pmod P$; by periodicity, s' and $s'+P$ are small elements of S_{kl}, and $s = s' + k((s'+P)-s')$ for some $k \in \mathbb{N}$.

Every $s \in S_{kl}$ is an integer combination of small elements of S_{kl} with coefficient sum 1. (5)

▶ **Lemma 14.** *Let $S_{kl} \subseteq \mathbb{N}$ be P-periodic from T, and let $B \triangleq T+2P$. For every strongly connected component C, every element of G_C is an integer linear combination of weights of closed walks inside C of length at most m all of whose chosen elements are at most B. In particular, if $G_C \not\subseteq \{0\}$, then C has a closed walk of length at most m whose weight lies in $[1, mB]$.*

Proof. Every closed walk decomposes, as a multiset of edges with their chosen elements, into simple cycles, so its weight is a sum of weights of simple cycles. A simple cycle inside C has length at most m. If a chosen element s is not small, write $s = s' + k(s''-s')$ with small s', s'' in the same edge set as in (5). Then the weight of the cycle equals $\gamma' + k(\gamma''-\gamma')$, where γ' and γ'' are the weights of the same cycle with s replaced by s' and s'', respectively. Repeating this for every edge of the cycle shows the first claim. For the second claim, if all closed walks of length at most m with small elements had weight 0, then so would every element of G_C; and such a closed walk has weight at most mB. ◀

▶ **Lemma 15.** *Let $S_{kl} \subseteq \mathbb{N}$ $(k,l \in [1,m])$ be P-periodic from T, and let $B \triangleq T+2P$. For each strongly connected component C with $G_C \not\subseteq \{0\}$, let $w_C \in G_C \setminus \{0\}$, and let $\Pi \triangleq P\prod_C w_C$. Then for all i, j, the set of weights of walks from i to j is Π-periodic from $c_m(B^2+\Pi)$, where c_m depends only on m.*

Proof. *Decomposition along components.* For a component C and $e, f \in C$, let $W_C(e,f)$ be the set of weights of walks from e to f inside C (including the empty walk if $e = f$). Every walk from i to j visits a sequence $C_1, \ldots, C_s$ of components along a path of the acyclic

component graph ($s \le m$), entering C_r at e_r and leaving it at f_r. So the set of weights of walks from i to j is the finite union, over such data, of the sums

$$W_{C_1}(e_1, f_1) + S_{f_1 e_2} + W_{C_2}(e_2, f_2) + \cdots + S_{f_{s-1} e_s} + W_{C_s}(e_s, f_s).$$

The edge sets are Π-periodic from T. We show below that each $W_C(e, f)$ is Π-periodic from $c'_m B^2$. Then, by Fact 2(2) applied to the at most $2m - 1$ summands, each sum, and hence the union, is Π-periodic from $2m(c'_m B^2 + \Pi)$.

Closed walks at e. Fix C and $e, f \in C$. Let $G_e \subseteq \mathbb{N}$ be the set of weights of closed walks at e inside C, including 0; it is a submonoid of $\mathbb{N}$, and $G_e \subseteq G_C \cup \{0\}$. Fix a walk from f to e inside C with small elements, of weight ρ, and a walk from e to f inside C with small elements, of weight w_0; then $\rho, w_0 \le mB$. Since $w + \rho \in G_e$ for every $w \in W_C(e, f)$, and $w_0 + G_e \subseteq W_C(e, f)$, we have

$$w_0 + G_e \subseteq W_C(e, f) \subseteq -\rho + G_e. \tag{6}$$

If $G_C \subseteq \{0\}$, then $G_e = \{0\}$ and $W_C(e, f) = \{w_0\}$ is Π-periodic from $mB + 1$. Otherwise, let $g \triangleq \gcd(G_C)$, which divides w_C and hence Π. For a closed walk inside C at a vertex v, of weight γ, we can go from e to v and back with small elements, with total weight $\pi_v \le 2mB$; both π_v and $\pi_v + \gamma$ lie in G_e. Let $A \subseteq G_e \setminus \{0\}$ be the finite set of the non-zero numbers π_v and $\pi_v + \gamma$, where γ ranges over weights of closed walks inside C at v of length at most m with small elements; all elements of A are at most $3mB$. Then $\gcd(A)$ divides each such $\gamma = (\pi_v + \gamma) - \pi_v$, hence divides g by Lemma 14; conversely g divides every element of G_e. So $\gcd(A) = g$. By the classical bound on the Frobenius number of numerical semigroups (e.g., $(\min A/g - 1)(\max A/g - 1)$), the monoid generated by A, and hence G_e, contains every multiple of g that is at least $(3mB)^2$. By (6), all elements of $W_C(e, f)$ are congruent to w_0 modulo g, and $W_C(e, f)$ contains every $n \equiv w_0 \pmod g$ with $n \ge mB + (3mB)^2$. Hence $W_C(e, f)$ is g-periodic, and thus Π-periodic, from $c'_m B^2$. ◀

C Proof of the closure lemma

Throughout this section, "for large a" means "for all sufficiently large a". We first note that parametric Presburger families can be glued along residue classes.

▶ **Lemma 17.** *Let $(X_a)_a$ be a family of subsets of $\mathbb{N}$ such that each single set X_a is Presburger definable, and let $M \in \mathbb{N}_+$. Suppose that for each residue r modulo M there is a parametric Presburger family $(X_a^r)_a$ such that $X_a = X_a^r$ for large $a \equiv r \pmod M$. Then $(X_a)_a$ is a parametric Presburger family.*

Proof. Let a_0 be a common bound beyond which the hypothesis holds, and let φ_r and $\psi_{a'}$ define X_a^r and $X_{a'}$ ($a' < a_0$), respectively. Then X_a is defined by

$$\bigvee_r \left(\exists k\ (Mk = a - r) \wedge a \ge a_0 \wedge \varphi_r\right) \vee \bigvee_{a' < a_0} \left(a = a' \wedge \psi_{a'}\right),$$

where $Mk = a - r$, $a \ge a_0$, and $a = a'$ are atomic formulas of the required form. ◀

We will apply Lemma 17 to sets of walk weights. For fixed a, each $S_{kl}(a)$ is defined by an ordinary Presburger formula, hence is semilinear and thus ultimately periodic. The construction in the proof of Lemma 16 below, carried out for this fixed a (where all parameters are constants, and bounded sets are finite), shows that the set of walk weights is Presburger definable. It therefore suffices to fix a residue r modulo M and to find a parametric Presburger definition valid for large $a \equiv r \pmod M$.

C.1 A normal form for polynomially bounded families

▶ **Lemma 18** (Simplex normal form). *Let $(X_a)_a$ be a parametric Presburger family with $X_a \subseteq [-Ca^D, Ca^D]$ for large a. Then there are finitely many tuples $(A_\nu, (W_{\nu j})_{j\in J_\nu}, \beta_\nu, L_\nu)$ with $A_\nu, W_{\nu j} \in \mathbb{Z}[a]$, $\beta_\nu \in \mathbb{Z}$, and $L_\nu \in \mathbb{N}_+$, such that for large a*

$$X_a = \bigcup_\nu X_\nu(a), \qquad X_\nu(a) \triangleq A_\nu(a) + \Big\{ \textstyle\sum_{j\in J_\nu} \lambda_j W_{\nu j}(a) \mid \lambda \in \mathbb{N}^{J_\nu},\ L_\nu \sum_j \lambda_j = a - \beta_\nu \Big\}.$$

Proof. Replacing X_a by $X_a + Ca^D \subseteq [0, 2Ca^D]$ (and A_ν by $A_\nu - Ca^D$ at the end), we may assume $X_a \subseteq [0, Ca^D]$. *Digits.* For $a > C$, every $n \in [0, Ca^D]$ has a unique expansion $n = \sum_{e=0}^{D} n_e a^e$ with $0 \le n_e < a$ for $e < D$ and $0 \le n_D \le C$. Let $Dig \subseteq \mathbb{N}^{D+2}$ be the set of tuples $(a, n_0, \dots, n_D)$ satisfying these bounds such that $\sum_e n_e a^e \in X_a$. We claim that, for some a_0, the set $Dig \cap \{a \ge a_0\}$ is definable in ordinary Presburger arithmetic, where a is treated as an ordinary variable. By [2, Thm. 3.7], X_a is defined by a formula in which every quantified variable is bounded by $0 \le y \le g(a)$ for a polynomial g. The free variable n is bounded by Ca^D as well. We expand every variable, free or bound, in base a as above (the free variable into $n_0, \dots, n_D$), as in [2, Lemma 4.5]. Every atomic formula becomes $\sum_s f_s a^s \le 0$, where each f_s is an integer linear combination of digits. As all digits are bounded by a, [2, Lemma 4.6] rewrites it, for large a, by guessing the carry h in $(h-1)a < f_0 \le ha$ among finitely many values, into a Boolean combination of ordinary Presburger inequalities in a and the digits and an inequality of the same kind with one power of a less. Since no variable is unbounded, iterating this eliminates all powers of a that multiply variables. The remaining inequalities without variables, of the form $g(a) \le 0$ with $g \in \mathbb{Z}[a]$, have a constant truth value for large a. Quantifiers over digits are bounded by $0 \le b < a$, which is an ordinary Presburger condition. This proves the claim.

Semilinear decomposition. By [7], $Dig \cap \{a \ge a_0\}$ is a finite union of linear sets $\{\beta + \sum_j \lambda_j \gamma_j \mid \lambda \in \mathbb{N}^J\}$ with $\beta, \gamma_j \in \mathbb{N}^{D+2}$ and $\gamma_j \ne 0$. Every γ_j has a positive a-coordinate $(\gamma_j)_a$: otherwise $\beta + \lambda\gamma_j$ would stay in Dig for all λ with a fixed a-coordinate and an unbounded digit, contradicting the bounds on digits. Let L be the lcm of all $(\gamma_j)_a$. Writing $\lambda_j = (L/(\gamma_j)_a)\lambda'_j + \rho_j$ with $0 \le \rho_j < L/(\gamma_j)_a$, each linear set is the finite union over ρ of the linear sets with base $\beta_\rho \triangleq \beta + \sum_j \rho_j \gamma_j$ and generators $\gamma'_j \triangleq (L/(\gamma_j)_a)\gamma_j$, each of which has a-coordinate exactly L. Hence an element of such a set has a-coordinate a iff $L \sum_j \lambda'_j = a - (\beta_\rho)_a$.

Back to integers. The map $(a, n_0, \dots, n_D) \mapsto \sum_e n_e a^e$ is linear in the digits for fixed a. So, with $A(a) \triangleq \sum_e (\beta_\rho)_{n_e} a^e$ and $W_j(a) \triangleq \sum_e (\gamma'_j)_{n_e} a^e$, the image of the linear set at parameter a is exactly the set $X_\nu(a)$ with $\beta_\nu = (\beta_\rho)_a$ and $L_\nu = L$. ◀

C.2 Iterated sums

For $X \subseteq \mathbb{N}$ and $k \in \mathbb{N}$, let $k \odot X \triangleq \{x_1 + \cdots + x_k \mid x_1, \dots, x_k \in X\}$, where $0 \odot X = \{0\}$.

▶ **Lemma 19.** *Let $X_\nu(a)$ be as in Lemma 18. Then the relation $\{(k, n) \mid n \in k \odot X_\nu(a)\}$ is a parametric Presburger family.*

Proof. We drop the index ν. If $a - \beta \notin L\mathbb{N}$, then $X(a) = \emptyset$, and $k \odot X(a)$ is $\{0\}$ for $k = 0$ and $\emptyset$ otherwise. If $a - \beta = LK$ with $K \in \mathbb{N}$, we claim

$$k \odot X(a) = \Big\{ kA(a) + \textstyle\sum_j \mu_j W_j(a) \mid \mu \in \mathbb{N}^J,\ \sum_j \mu_j = kK \Big\}.$$

The inclusion $\subseteq$ holds by adding the coefficient vectors of the k summands. For $\supseteq$, a vector $\mu \in \mathbb{N}^J$ with $\sum_j \mu_j = kK$ is the sum of k vectors in $\mathbb{N}^J$ with coordinate sum K each (remove

K units greedily, k times); this includes $k = 0$. Both cases are expressed by

$$(k = 0 \wedge n = 0) \vee$$
$$\exists K\, \exists \mu \left(LK = a - \beta \wedge n = k \cdot A(a) + \textstyle\sum_j \mu_j W_j(a) \wedge L \sum_j \mu_j = k \cdot a - \beta k \right),$$

in which only products of a variable with a polynomial in a occur. ◀

C.3 Walks

▶ **Lemma 16** (Closure lemma). *Let $(S_{kl}(a))_a$ for $k, l \in [1, m]$ be parametric Presburger families of subsets of $\mathbb{N}$. Assume there are $M \in \mathbb{N}_+$ and, for each residue r modulo M, polynomials $P_r, T_r \in \mathbb{Q}[a]$ such that for all sufficiently large $a \equiv r \pmod M$, every $S_{kl}(a)$ is $P_r(a)$-periodic from $T_r(a)$. Then for all i, j, the family of the sets of weights of walks from i to j (with edge weight sets $S_{kl}(a)$) is a parametric Presburger family.*

Proof. By Lemma 17, we fix a residue r modulo M and consider large $a \equiv r \pmod M$; we write $P \triangleq P_r$ and $T \triangleq T_r$. Each edge set decomposes as

$$S_{kl}(a) = F_{kl}(a) \cup (R_{kl}(a) + P(a)\mathbb{N}), \quad \text{where}$$
$$F_{kl}(a) \triangleq S_{kl}(a) \cap [0, T(a) - 1], \qquad R_{kl}(a) \triangleq S_{kl}(a) \cap [T(a), T(a) + P(a) - 1],$$

because $S_{kl}(a)$ is $P(a)$-periodic from $T(a)$. The families F_{kl} and R_{kl} are parametric Presburger families (multiply T and P by a common denominator) and polynomially bounded. By Lemma 18, they are finite unions of sets $X_\tau(a)$, where τ ranges over a finite index set Γ. We tag each $\tau \in \Gamma$ with its edge (k, l) and with its kind, *head* (from F_{kl}) or *tail* (from R_{kl}).

Consider the finite automaton over the alphabet Γ with states $[1, m]$ and a transition $k \xrightarrow{\tau} l$ for each τ tagged with (k, l). By Parikh's theorem [16], the set $\Pi_{ij} \subseteq \mathbb{N}^\Gamma$ of letter-count vectors of the words leading from i to j is semilinear, hence definable by a Presburger formula (without the parameter). We claim that n is the weight of a walk from i to j iff there are $(k_\tau)_\tau \in \Pi_{ij}$, $n_\tau \in k_\tau \odot X_\tau(a)$ for $\tau \in \Gamma$, and $y \in \mathbb{N}$ such that

$$n = \textstyle\sum_\tau n_\tau + P(a)y, \qquad \text{and } y = 0 \text{ if } k_\tau = 0 \text{ for every tail letter } \tau.$$

($\Rightarrow$) Each step of the walk uses an element of some $X_\tau(a)$ (head) or of some $X_\tau(a) + P(a)\mathbb{N}$ (tail). Collect the elements by letters, and let y be the sum of the multiples of $P(a)$ used in tail steps. ($\Leftarrow$) Take a word with letter counts $(k_\tau)_\tau$ from i to j, and distribute the k_τ summands of n_τ to the occurrences of τ. If $y > 0$, some tail letter occurs, and we add $P(a)y$ to one of its occurrences; this stays in $R_{kl}(a) + P(a)\mathbb{N} \subseteq S_{kl}(a)$.

By Lemma 19, the right-hand side is a parametric Presburger formula (after multiplying P by a common denominator). This proves the claim and the lemma. ◀

D Finite relational models

In this appendix we prove the relational part of Theorem 1.

▶ **Theorem 20.** *Let $h \in \mathbb{N}$ and let t be a term (over any variables) with $\mathrm{sh}(t) \le h$. Then there is a finite relational model in which Q_h and t differ. In particular, $\mathsf{REL} \not\models Q_h = t$.*

The structure of the proof is the same as for RSUB; we use finite cyclic versions of the chain models and describe only the differences. The cyclic setting is in fact simpler in the Kleene star case, since closed-walk weights then form a subgroup of a finite cyclic group.

D.1 Cyclic models

Let N be even and $a \in \mathbb{N}_+$. The *cyclic model* $\mathfrak{M}^\circ_a$ (with respect to N) is the relational valuation on the finite set $\Omega^\circ_N \triangleq \mathbb{Z}_N \times [1,m]$ with

$$z \mapsto \{((t,i),(t+k,i)) \mid k \in [0,N/2-1]\}, \qquad x \mapsto \{((t,i),(t+a+pi,i))\},$$
$$c \mapsto \{((t,i),(t+1,j))\}, \qquad y \mapsto \{((t,i),(t+pi-2,i))\},$$

where times are taken modulo N, and all other variables are mapped to $\emptyset$; we use the relational semantics $[\![-]\!]^{\mathsf{rel}}$. We call $W \triangleq [0,N/2-1]$ the *window*: z relates two points of the same phase whose time difference lies in W. The rotation $(t,i) \mapsto (t+s,i)$ is an automorphism, so with the difference sets $L^{t,\circ}_{ij}(a) \triangleq \{n \in \mathbb{Z}_N \mid ((0,i),(n,j)) \in [\![t]\!]^{\mathsf{rel}}_{\mathfrak{M}^\circ_a}\}$ we have $((t,i),(t',j)) \in [\![t]\!]^{\mathsf{rel}}_{\mathfrak{M}^\circ_a} \iff t'-t \in L^{t,\circ}_{ij}(a)$. The composition formula of Section 5.1 and (3) hold verbatim, with sums taken in $\mathbb{Z}_N$, and complements are taken in $\mathbb{Z}_N$.

D.2 The witnesses in cyclic models

We say that N is *admissible* for (a,h) if $f_r(a)+pi$ divides N and $2f_r(a)+2+pm < N/2$ for all $r \le h$ and $i \in [1,m]$.

▶ **Lemma 21.** *Let $L \triangleq \mathrm{lcm}(q_1,\dots,q_m)$, and assume that $L+pi$ divides N for all i and that $2L+2+pm < N/2$. If $L^{Q,\circ}_{ii}(a) \cap W = q_i\mathbb{N} \cap W$ for all i, then $L^{\Phi(Q),\circ}_{ii}(a) \cap W = (L+pi)\mathbb{N} \cap W$ for all i. Consequently, if N is admissible for (a,h), then $L^{Q_h,\circ}_{11}(a) \cap W = q_h(a)\mathbb{N} \cap W$.*

Proof. We follow the proof of Lemma 8. The term $z \cap \overline{1+c\,;c}$ relates the pairs in the same phase with time difference in $[3,N/2-1]$, and $z \cap \overline{Q}$ only depends on Q inside the window. For $n \in [3,N/2-1]$, the pair $((t,i),(t+n,i))$ is in $c\,;(z \cap \overline{Q})\,;c$ iff $L \nmid n-2$, since $n-2 \in W$. So $\mathsf{M}(Q)$ relates exactly the pairs $((t,i),(t+2+rL,i))$ with $r \ge 1$ and $2+rL < N/2$. The term $c\,;z$ relates exactly the pairs with time difference in $[1,N/2]$. For two $\mathsf{M}(Q)$-successors $v=(t+2+rL,i)$ and $v'=(t+2+r'L,i)$ of (t,i), the time difference from v' to v is $(r-r')L \in [1,N/2]$ if $r'<r$, and $N-(r'-r)L > N/2$ if $r'>r$, as $(r'-r)L < N/2$. Hence $\mathsf{J}(Q)$ relates exactly the pairs $((t,i),(t+L+2,i))$, and $\mathsf{J}(Q)\,;y$ those with time difference $L+pi$ in phase i. Its reflexive transitive closure relates time differences in the subgroup of $\mathbb{Z}_N$ generated by $L+pi$, which consists of the multiples of $L+pi$ as $L+pi$ divides N. The last claim follows by induction as in Corollary 9, starting from $Q_0 = x^*$ (note that $a+pi$ divides N). ◀

D.3 Periodicity relative to the boundary

Let $K \triangleq \{0,N/2\} \subseteq \mathbb{Z}_N$, a subgroup, and let $\delta(n)$ be the distance from $n \in \mathbb{Z}_N$ to K along the cycle; δ is invariant under translation by K. For $n \in \mathbb{Z}_N$ let $\sigma(n) \in \mathbb{Z}$ be the signed distance from n to the nearest point of K, so that $n \in \sigma(n)+K$. We say that $X \subseteq \mathbb{Z}_N$ is P-periodic from T if $n \in X \Leftrightarrow n+P \in X$ for every n such that $\delta(n+s) \ge T$ for all $s \in [0,P]$; that is, X is P-periodic on every stretch that keeps distance at least T from K. Complements, multiples of periods, and larger preperiods behave as before, and every subset of $\{n \mid \delta(n) < T\}$ is P-periodic from T.

▶ **Fact 22.** *Let $X,Y \subseteq \mathbb{Z}_N$.*

1. *Let X be P-periodic from T, and let $J=[l,l+q+P]$ with $\delta(n) \ge T$ for all $n \in J$. If $n \in X \Leftrightarrow q \mid n$ for all $n \in J$, then $q \mid P$.*
2. *If X and Y are P-periodic from T_X and T_Y, respectively, then $X+Y$ is P-periodic from T_X+T_Y+2P.*

Proof. (1) As in Fact 2. (2) Let $T \triangleq T_X + T_Y + 2P$, and let the stretch $[n, n+P]$ keep distance at least T from K. Every point $n + u - \varepsilon$ with $u \in [-P, 2P]$ and $|\varepsilon| < T_X$ is within distance $P + T_X$ of this stretch, so it has $\delta \geq T_Y + P$. ($\Rightarrow$) Let $n = \alpha + \beta$ with $\alpha \in X$, $\beta \in Y$. If the stretch $[\alpha, \alpha + P]$ keeps distance at least T_X from K, then $\alpha + P \in X$ and $n + P = (\alpha + P) + \beta$. Otherwise $\alpha + s = \kappa + \varepsilon$ for some $s \in [0, P]$, $\kappa \in K$, and $|\varepsilon| < T_X$. Then $\beta + u = n + s + u - \varepsilon - \kappa$, and since δ is invariant under K (here we use that K is a subgroup), every point of $[\beta, \beta + P]$ has $\delta \geq T_Y + P$. So $\beta + P \in Y$ and $n + P = \alpha + (\beta + P)$. ($\Leftarrow$) Symmetric, using the stretches $[\alpha - P, \alpha]$ and $[\beta - P, \beta]$. ◀

D.4 Upper bound in cyclic models

We now fix a polynomial $N \in \mathbb{Z}[a]$ taking even positive values, and consider the family $(\mathfrak{M}_a^\circ)_a$ with respect to $N(a)$; we represent elements of $\mathbb{Z}_{N(a)}$ by integers in $[0, N(a) - 1]$. Claim 13 makes sense verbatim with this notion of periodicity.

▶ **Proposition 23.** *If* $\deg N > d_t$*, then Claim 13 holds for* t *in the cyclic models with respect to* N*.*

Proof. We follow the induction of Section 5.1; the degree condition ensures that all preperiods of subterms are eventually much smaller than $N(a)$. Variables and constants have the difference sets $\emptyset$, $\{0\}$, $\{1\}$, $\{pi - 2\}$, $\{a + pi\}$, or W, all 1-periodic from $O(a)$. Complement, union, and composition are handled as before, using Fact 22 instead of Fact 2; definability is clear, as all sets are subsets of $[0, N(a) - 1]$ and reduction modulo $N(a)$ is expressed by $\exists k\,(n = n' - kN(a))$.

For the Kleene star, let $B \triangleq T_F + 2P_F$, and call an element s of an edge set small if $\delta(s) \leq B$. As in (5), if $\delta(s) > B$, write $s = \kappa + \sigma(s)$ with $\kappa \in K$; all points between $\kappa + T$ and s keep distance at least T from K, so s is an integer combination, with coefficient sum 1, of two small elements of the same edge set. Hence, as in Lemma 14, every element of G_C is an integer combination of weights of closed walks of length at most m with small elements. If all of them lie in the subgroup K, then $G_C \subseteq K$. Otherwise, one of them has weight $w = \sum_e (\kappa_e + \sigma_e) \notin K$ with $|\sigma_e| \leq B$, so $0 < |\sigma(w)| \leq mB$ (as $mB < N/4$). We therefore let $w_C(a)$ be the least non-zero $|\sigma(w)|$ over weights w of closed walks inside C of length at most m; it is a polynomial of degree at most d on each refined residue class by Theorem 12, and we set

$$P_{F^*} \triangleq P_F \cdot \prod_{C\,:\,G_C \not\subseteq K} 2w_C.$$

The degree bounds of Table 1 are unaffected by the factor 2.

It remains to show that the set of weights of walks from i to j is $P_{F^*}(a)$-periodic from $O(B^2 + P_{F^*}(a))$. We decompose walks along components as in the proof of Lemma 15, and keep its notation $W_C(e, f)$, G_e, w_0, ρ, and (6). If $G_C \subseteq K$, then $G_e \subseteq K$, and every element of $W_C(e, f)$ has $\delta \leq mB$, so $W_C(e, f)$ is periodic from $mB + 1$ with any period. Otherwise, G_e is a submonoid of the finite group $\mathbb{Z}_N$, hence a subgroup $g\mathbb{Z}_N$ with $g \mid N$, and (6) gives $W_C(e, f) = w_0 + G_e$. Let w be a closed walk weight at some $v \in C$ with $|\sigma(w)| = w_C(a)$. Going from e to v and back gives $\pi_v, \pi_v + w \in G_e$, so $w \in G_e$ and $2w = 2\,\sigma(w) \in G_e$, because $2K = \{0\}$. Hence g divides the integer $2w_C(a)$, and the coset $W_C(e, f)$ is $P_{F^*}(a)$-periodic everywhere. Fact 22(2) combines the at most $2m$ summands. (No Frobenius-type argument is needed here.)

Finally, definability follows as in Lemma 16, but more simply: the edge sets are bounded by $N(a)$, so by Lemma 18 they are finite unions of sets $X_\tau(a)$, and n is a walk weight iff

$0 \le n < N(a)$ and $n = \sum_{\tau} n_{\tau} - yN(a)$ for some $(k_{\tau})_{\tau}$ in the Parikh image, $n_{\tau} \in k_{\tau} \odot X_{\tau}(a)$, and $y \in \mathbb{Z}$. ◀

D.5 Proof of the theorem

We choose m and p as in Section 3, and let

$$N(a) \triangleq 2a^{d_t+1} \prod_{r=0}^{h} \prod_{i=1}^{m} \left(A_r(a) + pi\Delta^{\delta_r}\right) \in \mathbb{Z}[a].$$

Since $A_r(a) + pi\Delta^{\delta_r} = \Delta^{\delta_r}(f_r(a) + pi)$ for $a \in \mathcal{A}$, the number $N(a)$ is admissible for (a, h) for all large $a \in \mathcal{A}$. By Proposition 23, Claim 13 holds for t in the cyclic models with respect to N. Fix a residue class r modulo the modulus M that meets $\mathcal{A}$ infinitely often, and let P be the period polynomial and $T(a) = O(a^{d_t})$ the preperiod on it. Suppose that Q_h and t agree in all finite relational models, in particular in every $\mathfrak{M}_a^{\circ}$. By Lemma 21, $L_{11}^{t,\circ}(a) \cap W = q_h(a)\mathbb{N} \cap W$. The interval $J \triangleq [T(a), T(a) + q_h(a) + P(a)]$ lies in W and keeps distance at least $T(a)$ from K for large a, because $N(a)/2 \ge a^{d_t+1} q_h(a)$. So $q_h(a) \mid P(a)$ by Fact 22(1) for all large $a \in \mathcal{A}$ with $a \equiv r \pmod{M}$, and the Bézout argument of Section 3 yields a contradiction. ◀